

6 (a) Protein production in a prokaryote can be genetically controlled.

One example of the control of protein production in a prokaryote is the *lac* operon. The *lac* operon is functional when glucose is absent and lactose is present.

Complete Table 6.1, which shows information about the four genes associated with the *lac* operon.

Table 6.1

name of gene	type of gene	name of protein produced	function of protein produced
<i>lacI</i>		repressor protein	binds to operator to prevent RNA polymerase from binding to the promoter
<i>lacZ</i>	structural	β -galactosidase	
.....	structural	lactose permease	membrane transport protein to increase cellular uptake of lactose
<i>lacA</i>	structural	transacetylase	acetylates galactoside sugar residues to detoxify the cell



- (b) Insulin can be produced as a recombinant human protein using genetically engineered *Escherichia coli* bacteria. Recombinant insulin can be used to treat type 1 diabetes.

Describe the advantages of using recombinant insulin, instead of insulin from animal sources, to treat type 1 diabetes.

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